

1.

since the endpoints
are not included,

 $(-1, 1)$

$$-1 < w < 1$$

Ans: $\boxed{-1 < w < 1}$

2. (a)

$$64^{1/3} = 4$$

Ans: $\boxed{4}$

(b)
$$(-27)^{1/3} = 3$$

Ans: $\boxed{3}$

(c)
$$\left(-\frac{1}{27}\right)^{1/3} = -\frac{1}{3}$$

Ans: $\boxed{-\frac{1}{3}}$

3.
$$\begin{aligned} & (6x-9)(x^2-x+1) \\ &= 6x^3 - 6x^2 + 6x - 9x^2 + 9x - 9 \\ &= 6x^3 - 15x^2 + 15x - 9 \end{aligned}$$

Ans: $\boxed{6x^3 - 15x^2 + 15x - 9}$

4.

$$\frac{x^2 - 81}{x^2 - 25} \cdot \frac{x+5}{x+9}$$

$$= \frac{(x-9)(x+9)}{(x-5)(x+5)} \cdot \frac{(x+5)}{(x+9)}$$

$$= \frac{x-9}{x-5}$$

Ans:

$$\boxed{\frac{x-9}{x-5}}$$

5.

~~$\frac{9}{x-9}$~~ $\frac{9}{x-9} - \frac{18}{x^2} = 0$

$$9x^2 = 18(x-9)$$

$$x^2 = 2x - 18$$

$$x^2 - 2x + 18 = 0, \quad x = \frac{2 \pm \sqrt{4 + 72}}{2}$$

$$x = \frac{2 \pm \sqrt{76}}{2}$$

Ans:

$$\boxed{\frac{2 \pm \sqrt{76}}{2}}$$

$$6. \quad x(2x+5) \geq 0, \quad x=0, \quad x = -\frac{5}{2}$$

Testing intervals:

$$(-\infty, -\frac{5}{2}) \rightarrow +$$

$$(-\frac{5}{2}, 0) \rightarrow -$$

Include points

$$(0, \infty) \rightarrow +$$

Ans:

$$\boxed{(-\infty, -\frac{5}{2}] \cup [0, \infty)}$$

7. $\text{slope} = \frac{6}{7}$, $y\text{-intercept} = 5$
 $y = mx + b$, $y = \frac{6}{7}x + 5$

Ans: $\boxed{y = \frac{6}{7}x + 5}$

8.

$$f(x) = 5 - 4x^2$$

$$f(-2) = 5 - 4(4) = -11$$

$$f(0) = 5$$

Net change: $5 - (-11) = 16$

Average rate of change: $\frac{16}{0 - (-2)} = \frac{16}{2} = 8$

Ans: Net change = 16

Average rate = 8

9. $g(x) = x^2 - x - 20$
 $= (x-5)(x+4)$

Zeros: $x = 5, -4$

Vertex $x = \frac{1}{2}$

Looks like the graph in the $[-7, 7]$ window

Ans: $\boxed{[-7, 7] \text{ by } [-25, 20]}$

10. (a) $x^2 \rightarrow IV$

(b) $x^3 \rightarrow II$

~~(c) $\sqrt{x} \rightarrow I$~~ $\rightarrow$ (c) $\sqrt{x} \rightarrow I$, (d) $|x| \rightarrow III$

Ans: (a) IV

(b) II

(c) I (d) III

11.

$$P(x) = x^3 + x^2 - 20x$$

Factor out x : $x(x^2 + x - 20)$
 $= x(x+5)(x-4)$

Zeros: $x = -5, 0, 4$

Ans: $\boxed{-5, 0, 4}$

12. $A = Pe^{rt}$, $P = 1000$, $r = 0.065$

For $t = 3$: $A = 1000e^{0.195} \approx 1200$

For $t = 6$: ~~$A = 1000e^{0.39} \approx 1500$~~

$A = 1000e^{0.39} = 1500$

For $t = 18$: $A = 1000e^{1.17} = 3200$

Ans: (a) $\boxed{1200}$, (b) $\boxed{1500}$

(c) $\boxed{3200}$

13. Point: $(-\frac{3}{5}, -\frac{4}{5})$

$$\sin(t) = y = -\frac{4}{5}$$

$$\cos(t) = x = -\frac{3}{5}$$

$$\tan(t) = \frac{-4/5}{-3/5} = \frac{4}{3}$$

Ans: $\boxed{\sin(t) = -\frac{4}{5}}$, ~~cos~~

$\boxed{\cos(t) = -\frac{3}{5}}$, $\boxed{\tan(t) = \frac{4}{3}}$

14. Using Pythagorean theorem,

$$x^2 = 34^2 + 34^2, x = \sqrt{2312}$$

$$x \approx 48$$

Ans: $\boxed{48}$

15.
$$\frac{\cos(u) \sec(u)}{\tan(u)} = \frac{\cos(u) \cdot \frac{1}{\cos(u)}}{\tan(u)}$$

$$= \frac{1}{\tan(u)}$$

Ans: $\boxed{\frac{1}{\tan(u)}}$

6

$$16. \quad u = \langle 4, 1 \rangle, v = \langle 1, -3 \rangle, w = \langle 3, 2 \rangle$$

$$u \cdot v = 4(1) + 1(-3), \quad \cancel{u \cdot v} \\ = 1$$

$$\text{Ans: } \boxed{1}$$

$$17. \quad \sum_{k=6}^9 k(k+3)$$

$$\text{this becomes: } 6+7+8+9+3 = 33$$

$$\text{Ans: } \boxed{33}$$